MIL-STD-976 A
5 OCT 1981
SUPERSEDING
MIL-STD-976
31 AUGUST1977

MILITARY STANDARD

CERTIFICATION REQUIREMENTS
FOR
JAN MICROCIRCUITS

FSC 5962

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
Washington, DC 20301

Certification Requirements for JAN Microcircuits

MIL-STD-976

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/MSFC, EGO2, Marshall Space Flight Center, AL 35812, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

FOREWORD

The purpose of this standard is to establish criteria for certification (as required by MIL-M-38510). Definite criteria will assure that microcircuits are manufactured under conditions which have been demonstrated to be capable of continuously producing highly reliable products. This is accomplished by evaluating the manufacturer's capability for holding critical processes within established limits at specified critical points and continuously maintaining this capability during production. Certification and the maintenance thereof is a prerequisite to microcircuit qualification and is performed in advance and independent of procurement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

CONTENTS

			Page
Paragraph	1.	SCOPE	1
	1.1	Scope	1
	2.	REFERENCED DOCUMENTS	1
	3.	DEFINITIONS	2
	4.	GENERAL REQUIREMENTS	2
	4.1	Requirements	2
	4.1.1	Manufacturer pre-audit information	2
	4.1.2	Manufacturer audit	5
	4.1.3	Certification	6
	5.	DETAIL REQUIREMENTS	6
	5.1	System requirements	6
	5.2	Classes B and C line requirements	9
	5.3	Class S line requirements	12

FIGURES

Figure	1.	Typical process flow chart	3
Figure	2.	Example process control charts	7

TABLE

Table	I.	Substrate measurements	11

APPENDIXES

Appendix	A.	Cross Index of Calibration System Requirements	A-1
Appendix	B.	List of Qualifications Test Facilities or List of Quality Conformance Test Facilities	B-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard establishes the minimum requirements for the certification of manufacturing facilities/line(s) used in fabricating, assembling, and testing high reliability JAN microcircuits per MIL-M-38510. This includes plant facilities, equipment, personnel training, process controls, testing, and documentation.

2. REFERENCED DOCUMENTS

2.1 Government documents. The following documents form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510	Microcircuits, General Specification for

STANDARDS

FEDERAL

FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment

MILITARY

MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-977	Test Methods and Procedures for Microcircuit Line Certification
MIL-STD-45662	Calibration Systems Requirements

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquistion functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of certification audit shall apply.

DESC-EQM-42	Baseline Sheet JAN Microcircuits

(Application for copies should be addressed to Defense Electronics Supply Center, DESC-EQM, Dayton, Ohio 45444.

3. DEFINITIONS

3.1 Definitions. In addition to the definitions specified in MIL-M-38510, the following definitions apply:

* 3.1.1 Certification. The process by which a manufacturer's facilities/line(s) are audited and approved for compliance with the requirements of this standard and MIL-M-38510 prior to and independent of procurement.

* 3.1.2 Certification team. The Government qualifying activity's designated representative(s) charged with the responsibility of auditing a manufacturer's facility/line.

* 3.1.3 Facility. Equipment, documentation, and personnel used to support the manufacture, assembly, and testing of microcircuits (not necessarily part of a line).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

* 3.1.4 Microcircuit technology. A group of microcircuits which are manufactured on the same line using the same materials, procedures, and similar equipment, such as TTL, CMOS, NMOS, etc. See Appendix E, MIL-M-38510.

* 3.1.5 Line. A sequence of process steps, including necessary equipment, facilities, and personnel, used to manufacture microcircuits.

4. GENERAL REQUIREMENTS

4.1 Requirements. The following requirements are applicable for the certification of product assurance levels for JAN microcircuits as specified in MIL-M-38510. The certification team will determine the adequacy and compliance to the requirements and shall report their findings and recommendations to the qualifying activity.

* 4.1.1 Pre-audit information submission. The manufacturer shall submit to the qualifying activity the information required in 4.1.1.1 through 4.1.1.3 for review and approval prior to the initial audit. The manufacturer's flow chart for each wafer fabrication line to be certified must comply with requirements of MIL-M-38510, paragraph 20.1.3.2. At least one JAN product which is intended for manufacture on each wafer fabrication line to be certified for each particular process shall be identified.

* 4.1.1.1 Product assurance program plan. The plan shall be in accordance with and cross-referenced to the applicable paragraphs of MIL-M-38510, Appendix A. The manufacturer's flow chart for the line must be complete, current, and accurate and shall contain, as a minimum, the type of information as shown in the example in figure 1 for all processes to be certified.

4.1.1.2 Calibration information.

4.1.1.2.1 Calibration system description. The system description shall be in accordance with MIL-STD-45662 and cross-referenced to MIL-STD-45662 requirements using Appendix A herein. This is to assist the manufacturer in speedily locating the pertinent documentation during audits of his facility by the certification team.

* 4.1.1.2.2 Calibration standards list. The calibration standards list shall contain all primary standards and precision measurement equipment used as reference or transfer standards in the in-plant calibration program. (Certificates showing traceability to the National Bureau of Standards (NBS) are to be maintained current at the manufacturer's plant and available at the time of audit.) The list of calibration standards shall be itemized using the DESC form 36 in Appendix B or in any other format containing the equivalent information.

4.1.1.2.3 Calibration laboratories list. The list shall include names and addresses of all calibration laboratories utilized. The test laboratory must verify that the calibration laboratories utilized are, in fact, capable of performing the required services. (See 5.7.1 of MIL-STD-45662.)

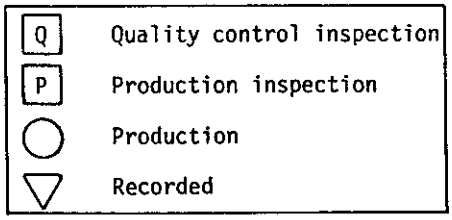
4.1.1.3 Internal visual documentation. The manufacturer's inspection criteria shall be cross-referenced to the appropriate test condition of Method 2010, MIL-STD-883. Also, the means for implementing the requirements of 3.7.1.1 of MIL-M-38510 are to be indicated. If the quality control personnel use a different document(s) than the manufacturing personnel, both sets of documents shall be cross-referenced.

4.1.1.4 Testing information. The following test information shall be made available to the certification audit team:

4.1.1.4.1 Device type information. The manufacturer's part number and Military part number(s) (if available) of the device type(s) to be qualified are to be listed (per the detail Military Specification). If the detail Military Specification is not in final form, this information shall be supplied as called out in the proposed or initial draft of the detail Military Specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

PRODUCT FLOW	OPERATION TITLE	SPECIFICATION NO.	REV & DATE (See note 1)
Q	Substrate inventory	M & SS 341, Q & R 3540	
P	Issue material/lot form.	LIN 4540	
	Wafer clean	LIN 4541	
	P-oxidation	LIN 4503	
P	Oxide X_J evaluation	LIN 4504	
P	Diffusion surface inspec.	LIN 4520	
Q	Resist	M & SS 401	
	Apply photoresist	LIN 5504	
	Soft bake	LIN 5505	
Q	Mask	Q & R 3560	
	Aline & expose photores.	LIN 5506	
	Develop	LIN 5507	
	Hard bake	LIN 5540	
P	Develop inspect	LIN 7820	
	Backspin resist	LIN 5531	
	Hard bake	LIN 5540	
P Q	Develop inspect	LIN 7820, Q & R 6555, Q & R 6570	



NOTES:
1. Specification revisions and dates must be current at the time of audit. This information need not be placed on the flow chart. However, this information must be made available to the certification team during the audit.
2. This flow chart is not complete and is used as an example to show the type of information which shall be included.

FIGURE 1. Typical process flow chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

* 4.1.1.4.2 Test facilities list. The test facilities list shall include all test and measurement equipment which is to be used for qualification testing under the Detail Military Specification. All the equipment necessary to perform each test shall be itemized in the DESC form 36, as shown in Appendix B, or in any other format containing the equivalent information. It is recommended that no more than one test method (except burn-in/life and electrical test methods) be listed on any one page to facilitate any equipment change approvals. Test facilities lists require each item of test equipment to be cross-referenced to the applicable qualification test paragraph of the specification. As nearly as practicable, equipment used collectively to perform the tests, or make certain measurements, will be grouped together on the lists. Meters that are not an integral part of the test equipment may be listed separately. Where tests are repeated in ensuring subgroups of the qualification table, the complete relisting of test equipment for the test is not necessary. An annotation, "See (applicable paragraph) above' will suffice.

* 4.1.1.4.3 Moisture chart. If a moisture chamber is included in the test facilities list, a completed humidity chart (wet and dry bulb readings or differential reading) demonstrating compliance with Method 1004, MIL-STD-883, shall be submitted. The chart must also include the name of the manufacturer of the chamber, the model and serial number, and the date of recording. During the trial run, which is necessary in order to obtain a recording on the chart, the chamber must be loaded with a mass of metal approximately equal to the mass of the maximum number of samples to be tested at one time. This procedure and the data recorded are to approximate actual testing conditions. For example, when the subcycle of step 7 in Method 1004, MIL-STD-883 (-10°C, 3 hours) is not to be performed in the same chamber, the removal and replacement of the test mass of metal must be noted on the chart.

* 4.1.1.4.4 Shock photo. If shock test equipment is included in the test facilities list, a photograph of the calibration wave must be available at the time of the audit. The manufacturer shall demonstrate compliance with the applicable test condition of Method 2002, MIL-STD-883. The photograph is to be dated and the axes properly labeled, showing the unit of measure and scale.

* 4.1.1.4.5 Constant acceleration information list. This list is to indicate the package types for which approval is required.

* 4.1.1.4.6 Testing flow charts. Flow charts or shop travelers for testing the parts shall be available at the time of the audit.

4.1.1.5 Verification procedure documentation. The manufacturer shall supply the procedures which are used internally to verify that his ac/dc test programs, switching and special schematics, and burn-in schematics are in accordance with the specification requirements. Examples of typical schematics and test programs shall also be supplied for each technology within which qualification is planned.

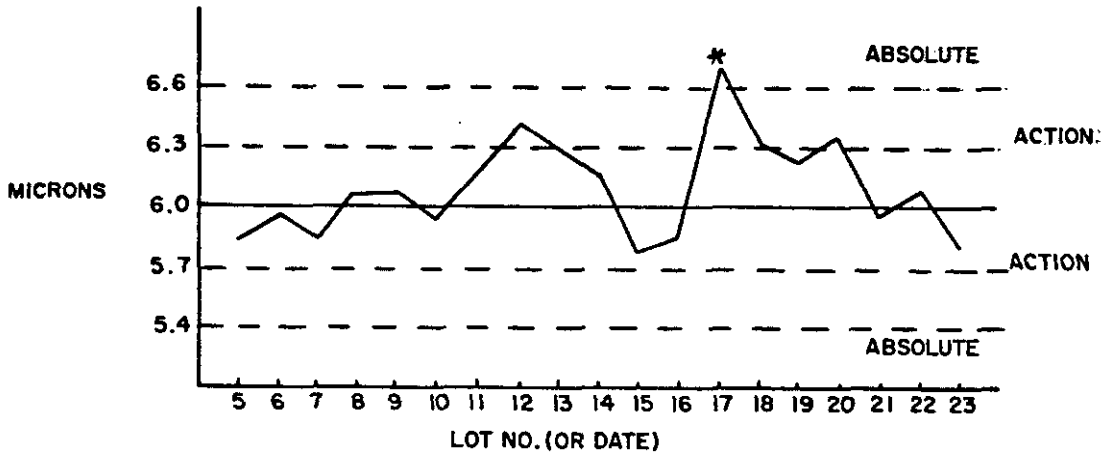
* 4.1.1.6 Package type. Samples of the package types (one sealed and one unassembled set) (see definitions in MIL-M-38510) which are intended for production on the line to be certified shall be available at the time of the audit.

* 4.1.1.7 Audit baseline sheet. DESC form EQM 42 Sections I and II shall be current and complete for each line and technology for which the manufacturer is seeking certification. This shall be verified prior to the certification audit.

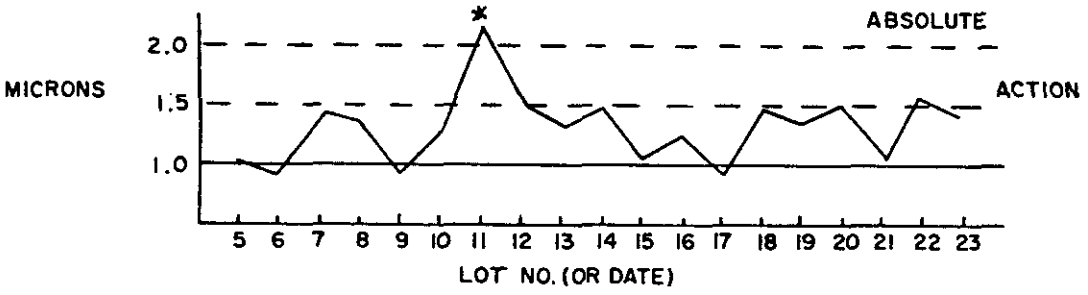
4.1.2 Manufacturer audit. The certification team will audit the manufacturer's facilities/line(s). The purpose of the audit is to determine that the controls imposed on manufacturing, inspection, and testing of JAN microcircuits are sufficient to assure conformance with the requirements of MIL-M-38510 and this standard.

4.1.2.1 Capability demonstration. Evidence shall be shown to the certification team during the audit that the certification requirements herein are being regularly met in production and further research or development is not required. The manufacturer shall provide (or make available for review during the audit of each line) representative samples, data and documentation as required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.



a. Mean ($\overline{X}$) Epitaxy thickness (acceptance criteria 6 ±0.6 microns)



b. Range, epitaxy thickness

*Corrective action report filed and disposition of product documented

FIGURE 2. Example process control charts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

4.1.2.2 Audit report. The qualifying activity will report the results of the audit to the manufacturer.

* 4.1.3 Certification. Certification is normally valid for a period of 2 years and can be extended or terminated only by written notification from the qualifying activity. During the duration of the certification, the manufacturer is subject to a periodic review by the qualifying activity or its representatives. Certification will not normally be extended beyond the 2-year period without a re-audit of the manufacturer's facility for compliance with this standard and MIL-M-38510.

* 4.1.3.1 Loss of certification. The manufacturer may lose certification for failing to comply with the requirements of MIL-M-38510 or this standard.

* 4.1.3.2 Reinstatement of certification. Manufacturers may have certification reinstated by successfully completing all the requirements referred to in MIL-M-38510 and this standard, or such portions thereof as the qualifying activity may direct in order to remedy the deficiencies leading to the loss.

5. DETAIL REQUIREMENTS

5.1 System requirements. In addition to section 4, the manufacturer shall document, control, and meet the following:

5.1.1 *Documentation. Documents shall* be maintained (in accordance with Appendix A of MIL-M-38510) which specify materials, calibration techniques, processing, test and measurements, controls and procedures. These documents shall cover all process steps to be certified. The documentation shall be available to the operating personnel at all times. It shall be made available to the certification team for the purpose of *verifying its existence, coverage,* implementation and adequacy.

* 5.1.1.1 Lot traveler. A lot traveler shall be used for each lot. The lot traveler shall include as a minimum: lot identification, operation, quantity, date of operation, and operator identification.

5.1.1.1.1 Wafer lot acceptance for Classes. A form which contains the data required by Method 5007, MIL-STD-883, shall be utilized for each wafer lot.

* 5.1.1.2 Process control charts. Process control charts (see figure 2) shall be maintained during the manufacturer of microcircuits. The charts or logs shall contain information such as: process step, lot number *and/or* date, *mean, action* limits and absolute limits and range. Where absolute limits are exceeded, the manufacturer shall document the corrective action taken.

* 5.1.1.3 Equipment maintenance. Where appropriate (e.g., ion implanter), the procedures used, the records kept, *and the nature of preventive and repair* maintenance shall be specified in accordance with 20.1.1.9 and 20.1.2.5 of MIL-M-38510, Appendix A.

* 5.1.1.4 Flow charts for testing. A flow chart for testing (for screening per Method 5004, MIL-STD-883, and for groups A, B, C, and D testing per Method 5005, MIL-STD-883) shall be documented, have a document number, and be controlled such that change of flow and date of flow change can be determined. (See figure 1).

5.1.2 Incoming inspection. Methods and procedures which are used to control inspection, storage, and handling of incoming materials shall be documented. Records shall be provided which verify that materials used in production meet the requirements of the manufacturer's specifications and of the general and detail MIL-M-38510 specifications. As an example, the following items will be covered:

5.1.2.1 Substrate. Thickness, flatness, surface orientation, conductivity type, resistivity, visual examination.

5.1.2.2 Wire. Composition, uniformity, hardness, diameter, elongation, tensile strength, purity, cleaniness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

5.1.2.3 Metal. Composition, purity.

* 5.1.2.4 Package material and lead frames. Dimensions, composition, integrity test, purity, plating, underplating, and tolerances.

5.1.2.5 Chemical, photoresists, and gases. Composition, purity, grades.

* 5.1.2.6 Masks. Geometry (lines, gaps, and tolerances), pinholes (density, size and distribution), scratches, edge raggedness.

5.1.3 Calibration. Calibration shall be in accordance with MIL-STD-45662.

5.1.4 Environmental control. Specify, control and record the relative humidity, temperature and particle count for each critical process step (e.g., wafer fabrication, assembly). The procedures and techniques for measuring these environmental parameters and limits shall be documented. The procedures shall contain corrective actions for out-of-tolerance environmental conditions. Particle counts shall be in accordance with Federal Standard 209.

* 5.1.5 Water control. Specify, measure, control and record the purity of water in terms of: minimum resistivity (action and absolute limits) at 25°C, maximum total solids, maximum organic impurtities, maximum bacteria count, and maximum chlorine count.

5.1.6 Electrical testing for JAN microcircuits.

5.1.6.1 Hardware. All electrical test equipment and fixtures used for testing JAN microcircuits shall be suitable to meet applicable test conditions. Set up and operation procedures for each operation/equipment shall be made available for determination of adequacy.

5.1.6.2 Programs and tapes. Test programs and corresponding tapes shall meet the requirements of the detail microcircuit specifications. A system shall be documented which insures that, if a specification is revised, the corresponding program(s) and tape(s) are also changed, and that the correct program is used.

5.1.7 Environmental testing for JAN microcircuits. All environmental test equipment and fixtures used for testing JAN microcircuits shall be suitable to meet applicable test conditions. Set up and operation procedures for each operation/-equipment shall be made available for determination of adequacy.

5.1.8 Failure analysis. Failure analysis and corrective actions shall be performed and documented in accordance with MIL-M-38510.

* 5.1.9 Handling of wafers, substrates and tools. Provisions shall be made for the careful handling of wafers, substrates, tools, fixtures, etc., used in the production cycle to prevent damage and contamination.

5.1.10 Training. Training shall be in accordance with 20.1.1.2 and 20.1.2.1 of MIL-M-38510, Appendix A. Some prime examples of critical areas for training are wafer fabrication, wire bonding, die mounting, internal visual inspection and all quality control operations.

5.2 Classes B and C line requirements. The following processes shall be documented and controlled (including accept/reject criteria) and are applicable to the certification of a line on which the manufacturer intends to produce Class B and Class C JAN microcircuits. In addition, the manufacturer must meet the requirements of sections 4 and 5.1 of this standard.

* 5.2.1 Oxidation and deposition. Each combination of materials (SiO_2, Si_3N_4, Al_2O_3) formed on the substrate or wafer shall be identified and the methods of formation (thermal oxidation, sputtering) shall be specified. Equipment, thickness, and tolerances shall be specified and controlled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

5.2.2 Patterning. Techniques (photolithography, beam) and materials used to form patterns shall be specified.

* 5.2.2.1 Photoresist. The following shall be specified: preparation and evaluation (specific gravity, viscosity, solids residue, definition of line width, pinhole count), storage conditions (temperature, time, type of container), application (humidity control, rpm of spinner, acceleration, spin time), pre-bake and bake (time, temperature, environment), development (cycle times, temperature, chemicals) and removal. If the photoresist thickness is measured, rpm of spinner, acceleration and spin time need not be specified.

* 5.2.2.2 Etching. The controls and tolerances on each patterning process shall be specified for each technique (i.e., wet chemical etch, dry plasma etch, etc.). For wet etch methods, the etchant information shall include: composition, grade, concentration, frequency of etchant replacement, temperature, etch rate and time of each oxide or metal layer agitation, and method of drying.

5.2.2.3 Alignment and exposure. The following shall be specified: exposure (uniformity, contact pressure, wavelength, light intensity, time, environment), and visual inspection (magnification, lighting, visual aids).

5.2.3 Epitaxy. Techniques, controls, materials and equipment used in forming an epitaxial layer on a substrate shall be specified. This shall include temperature, cycle times, sheet resistivity of epitaxial layer, thickness of epitaxial layer, gas flow-rates, loading and unloading of wafers (dust removal, wafer placement on susceptor), wafer precleaning, susceptor and tube cleaning. Instruments used for temperature measurement shall be calibrated.

5.2.4 Junction formation. Techniques and materials used in forming junctions shall be identified.

5.2.4.1 Diffusion. Profiling measurements, temperature measurements, diffusion tube and associated handling hardware cleaning frequency and techniques, temperature tolerance, flow rate of gasses, wafer loading and unloading, storage, calibration of thermocouples, and dopant source (grade, type, etc.) shall be specified.

5.2.4.2 Ion implantation. Dopant source (grade, type, etc.), calibration frequency of integrated dose rate measuring equipment, power supplies and vacuum equipment, frequency and cleaning technique of the entire system, wafer loading and unloading, and maintenance records shall be specified.

5.2.4.3 Test on junctions. Test results shall be made available on critical parameters: Beta, breakdown voltages, V/I, leakage current, and others as specified. These results should show the action and absolute limits, frequency of test, and the type of record.

5.2.5 Metallization. Equipment for depositing metal, each composition of metal (Al, Ti-W-Au, Pt-Ti-Pt-Au, etc.) method (vacuum chamber electroplating, sputtering, etc.), heat source (E-beam, filament, etc.), temperature of substrate during deposition, metal thickness and tolerance, deposition rate of metal, motion (stationary, rotary, or planetary), cleaning of chamber, thickness control, weight of melt, and sintering (time, temperature and ambient gas) shall be specified.

5.2.6 CV plotting. When used, the techniques, equipment, frequency of test, test conditions (time, temperature, bias, etc.), and action and absolute limits shall be specified. CV plotting is required on all MOS lines and certain linear lines.

5.2.7 Glassivation. When used, the method of formation (oxidation of silicon, quartz sputtering, etc.) and the materials used shall be identified. The controls for the glassivation layer(s) shall be documented and include temperature, time, thickness, flow rate of gasses, motion of wafer during deposition, control of power and frequency, and dopant concentration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

5.2.8 Final wafer thinning. Techniques, methods of removal (lapping, etching, etc.), and materials used in reducing wafer thickness shall be specified. Pressure, grit size, speed of rotation, removal of coating, cleaning, front side protection, parallelism, final thickness, tolerances and other pertinent parameters shall also be specified.

5.2.9 Scribing. Each method (diamond, laser, etch, etc.) of scribing shall be specified.

5.2.9.1 Diamond. The type of machine, scribing tool, tool point (e.g., heel), tool angle, tool pressure, speed and direction of tool travel, depth of scribe line, orientation of die pattern, use of oxide and metal free channels, and debris removal shall be specified.

5.2.9.2 Laser. The type of equipment, power, beam resolution, depth of scribe line, cleaning, protective overcoat and pulse rate shall be specified.

5.2.9.3 Etching. The concentration, frequency of replacement, temperature and time of etch shall be specified.

* 5.2.10 Dice separation. Each method of dice separation to be certified shall be specified.

* 5.2.10.1 Manual. Wafer protection, roller diameter, composition, and cleanliness shall be specified.

* 5.2.10.2 Sawing. The wafer feed rate, slurry, washing, and kerf depth shall be specified.

* 5.2.10.3 Other. Wafer protection, anvil, pressure, etc., shall be specified as applicable. The method of storing sorted dice shall be specified.

* 5.2.11 Die mounting. For each type of wafer material (Si, GaAs, sapphire, etc.), die mounting material (gold, epoxy, etc.), package mounting material (Au-Kovar, etc.) and mounting configuration shall be specified. Time, temperature, power, pressure, scrubbing, cleanliness, and environment shall also be specified.

* 5.2.12 Interconnect bonding. Techniques (ultrasonic, thermocompression, etc.), material (Al, Au, etc.), material diameter, and kind of lead (wire, beam lead, etc.) used in connecting die to the package shall be identified. The following shall also be specified: temperature, pressure dwell time, condition of capillary or electrode control, ultrasonic power, loop height, thickness/width ratio of bond, and environments.

5.2.13 Internal visual inspection. The circuits shall be visually inspected utilizing adequate visual aids, inspection criteria, storage, and equipment. The inspection criteria and handling and storage between internal visual and seal shall be in accordance with Method 2010, condition B of MIL-STD-883.

5.2.14 Sealing. The type of material and seals (solder, devitrified glass) used in production of packages (TO-5, flat-pack, dual-in-line, etc.) and the method (beam welding, resistance welding, belt furnace, etc.) shall be specified. The following shall also be specified: pre-seal bake (time, temperature, atmospheric conditions), heat or power, humidity during sealing (specify moisture content in ppm), flow rate of gasses, profile and welding controls (pressure, power, time), cleaning, and package orientation.

* 5.2.15 Marking. The procedures and materials used for marking shall be specified.

5.3 Class S line requirements. In addition to sections 4, 5.1, and 5.2, the manufacturer shall document and demonstrate capability to control the following:

5.3.1 Substrate measurements. Using documented techniques and existing equipment, the manufacturer shall have performed and continue to be capable of performing the measurements specified in table I. If the certification team has reason to doubt the capability of the manufacturer to correctly make these measurements, the team may request that additional measurements be made in their presence. The team may also

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

request that the substrates, including the recorded measurements, be submitted to the qualifying activity for performance testing in accordance with some or all the measurements specified in MIL-STD-977 and referenced in table I.

* 5.3.2 Pinhole and crack measurements. Demonstrate the method used during production for detecting and measuring the densities and size of pinholes and cracks in a typical oxide layer. Test Methods 3040 and 3050 of MIL-STD-977 or an equivalent test shall be used. The certification team may request that samples of oxides that have these imperfections identified and measured be submitted for verification in accordance with the above test methods. The wafer lot reject limits, disposition of reject wafers, and corrective actions shall be specified.

* 5.3.3 Stability of devices. To assure the stability of each process step in producing devices, the manufacturer shall have documented procedures to show that he has developed and is using tests such as high temperature reverse bias, BV_{CEO}, h_{FE}, I_{CBO}, I_{EBO}, V_T, BV_{DSS}, BV_{GO}, sheet resistivity, or others. The manufacturer may be required to demonstrate these tests to the certification team. Records, such as X, R charts shall be available with parameter limits that show control of the process used for producing devices.

5.3.4 CV plotting. Current CV plots shall be made available to the certification team, and CV plotting may be required to be performed during the audit (using Method 2500 of MIL-STD-977 or equivalent). Records such as X, R charts or data logs shall be made available with parameter limits that show continuous control of the process. The CV plot is mandatory for bipolar devices which operate above 10 volts and for MOS devices. The V_T or V_{FB} shall not exceed the limit establish for that particular technology. For MOS devices and bipolar devices which operate above 10 volts, CV plots are to be performed on at least each lot of devices with appropriate data recording and retention.

*TABLE I. Substrate measurements.

Test	MIL-STD-977 Test Method No.	Required Max tolerance
1. Thickness	1580	± 1 mil of mean
2. Flatness (Bowing or Waviness)	1580	For 2" wafers: 1.5 mils max For 3" wafers: 2.0 mils max For 4" wafers: 3.0 mils max For 5" wafers: 4.0 mils max For 6" wafers: 5.0 mils max
3. Parallelism	1580	For 2" wafers: 0.5 mils max For 3" wafers: 1.0 mils max For 4" wafers: 1.5 mils max For 5" wafers: 2.0 mils max For 6" wafers: 2.5 mils max
4. Surface Roughness Surface Roughness	1600 1600	500A max No visible roughness
5. Crystal Perfection	1550, 1560	
a. Dislocation Count		500 pits/cm^2
b. Lineage		1/cm^2
c. Slip		1/cm^2
6. Surface Orientation	1530	± 1/2° of specification
7. Conductivity Type (lowest resistivity)	1570	n or p
8. Resistivity	1510	Within specification

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

5.3.5 Induced wafer defects. When requested by the certification team, the manufacturer shall demonstrate that the quantity of induced dislocations and other defects are not sufficient to be deleterious to the characteristics of the junctions, oxide layers or other structures built in or on the substrate.

* 5.3.6 Photoresist pinholes. The test method and the maximum density of pinholes shall be specified for each photoresist type.

5.3.7 Masks. The percent defects allowable on working masks shall be specified.

5.3.8 Epitaxy. The stacking fault count allowable of the epi-layer shall be specified and recorded.

* 5.3.9 Stability of metallization. The tests specified in paragraphs 5.3.9.1 and 5.3.9.2 are applicable to manufacturing lines on which devices having any of the following are manufactured:

a. Metal line widths narrower than 2 micrometers

b. Multilevel structures, where one of the levels is other than pure or doped aluminum or gold (i.e., polysilicon, doped silicon, etc.)

c. Metals other than aluminum or gold

d. LSI

As an alternate to the requirements of paragraphs 5.3.9.1 and 5.3.9.2, data may be presented to the certification team which demonstrates the stability of the metallization and contacts of actual devices from the lines to be certified under temperature and high current density stresses. The certification team shall evaluate the manufacturer's test procedures and results for acceptability.

* 5.3.9.1 Effect of time and temperature. Data shall be presented to the certification team for each metallization type used on the line(s) to be certified and to demonstrate the stability of contact resistance under accelerated time and temperature conditions. If more than one conductor-contact type is used (e.g., metal-polysilicon, metal-multilayer, polysilicon-silicon, etc.), data shall be presented for each conductor-contact type.

An acceptable test would consist of subjecting the conductor-contact test patterns to a storage temperature of 200°C for 2000 hours. Each test pattern should contain at least 60 conductor-contacts of each type, unless otherwise specified.

Five test patterns or actual devices from each of five differenct deposition runs shall be used for the test. The contact resistance shall be measured before and after the temperature stress test and the data shall be recorded. The manufacturer's test plan, procedures, failure criteria and test results shall be reviewed for acceptablity by the certification team.

Equivalent time/temperature tests are allowed when approved by the certification team. In particular, the Class S time-temperature regression curve of Method 1005 of MIL-STD-883 may be used to determine an equivalent test.

* 5.3.9.2 Effect of current. Data shall be presented to the certification team for each conductor type and type of deposition system to be certified to demonstrate absence of electro-migration under current and temperature accelerated test conditions.

Test patterns may be used which contain metallization crossovers, contacts, etc., which are laid out using design rules (layouts, dimensions, etc.) representative of the worst case design rules used in the production line to be certified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

Five test patterns or actual devices containing each conductor type from five different deposition runs for each metallization equipment type (E-beam, filament, etc.) on the line to be certified shall be tested and the data recorded for review by the certification team. The manufacturer's test plan, procedures, failure criteria and test results shall be reviewed for acceptability by the certification team.

Equivalent time/temperature tests are allowed when approved by the certification team. In particular, the Class S time-temperature regression curve of Method 1005 of MIL-STD-883 may be used to determine an equivalent test.

The audit team will review the data generated by the test. The data shall support a minimum activation energy of one electron-volt, unless otherwise approved by the team.

* 5.3.10 Inspection by scanning electron microscope (SEM). A continuing SEM program shall be established to insure control over metallization at oxide window ledges. SEM photographs shall be presented demonstrating capability of meeting the magnification requirements of Method 2018 of MIL-STD-883. The angles from which the photographs were taken shall be varying, and they shall include circuits manufactured over a period of at least the preceding 6 months. These photos should include, as a minimum, views of the deepest oxide step covered by metallization lines. These photographs should confirm that the metallization process meets the SEM requirements of MIL-STD-883, Method 2018.

5.3.11 Control of materials for assembly area. The criteria, sampling requirements and acceptable control limits and handling which reflect the higher reliability requirements of Class S shall be identified. As a minimum, these requirements shall include:

a. Formation of special inspection lots for critical materials and for packages.

b. Sample assembly of critical items into packages and their acceptability.

5.3.12 Effect of time and temperature on electrical resistance and strength of bonds. Evidence shall be presented to demonstrate the mechanical and electrical integrity of the bonds with respect to such factors as (1) flexing of bond wire due to thermal expansion and (2) microcracks or microvoids at the wafer interface.

* 5.3.13 Die shear strength. The die shear strength test shall be performed at the start and finish of operator change, package type change and die size change. A die shear strength test in accordance with Method 2019 of MIL-STD-883 shall be conducted on a minimum of two devices from each die attach station at the start of each shift and after every two hours of production. In the event that the die shear is less than the value of figure 4 of Method 2019, the die attach station shall be closed down until tests show that satisfactory operation has been reestablished. A procedure for the traceability, recovery, and disposition of all units die-attached since the last successful die shear test is required. This procedure shall provide for sample size, reject criteria and disposition of failed lots. This test may be conducted on the same samples used for the wire bond strength test.

* 5.3.14 Wire bond strength test. The manufacturer's wire bond strength test shall be conducted on each sample in accordance with the requirements of MIL-STD-883, Method 2011, test condition D. The manufacturer shall submit the sampling plan to the team for approval, which will include start and completion of shifts, frequency of sampling during the shift, change of operators, spools, packages, wire size, lot size and other related factors (e.g., bond pull testing 2 devices approximately each two hours of production). Pull strength data shall be read and recorded and maintained in accordance with the specified requirements. Data shall include the force, in grams, required for failure, the physical location of the point of failure, and the nature of the failure. In the event that any bond strength is less than the pre-seal value given in Table I, Method 2011 of MIL-STD-883, the bonder shall be inactivated immediately and not returned to production until tests show that satisfactory operation has been reestablished. A procedure for the traceability, recovery, and disposition of all units bonded since the last successful bond strength test is required. This procedure shall provide for sample size, number of bonds/device to be tested, reject criteria and disposition of failed lots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

* 5.3.15 Internal visual inspection. The circuits shall be visually inspected utilizing adequate visual aids, inspection criteria, storage, and equipment. The inspection criteria and handling and storage between internal visual and seal shall be documented by the company and shall be in accordance with either Method 2010 (condition A), or Method 2017 of MIL-STD-883.

* 5.3.16 Particle contamination. The manufacturer shall have the capability to perform particle impact noise detection (PIND) testing in accordance with Method 2020 of MIL-STD-883. Records of PIND testing shall be made available to the team for review. Data for a three month period shall be available.

* 5.3.17 Internal water vapor levels. Documentation from a certified test laboratory shall be provided at the time of audit showing the actual internal moisture level of devices for a package type for which qualification is desired. These devices must be processed and screened to the Class S baseline and the internal moisture level for the package shall not exceed 5000 ppm at 100°C.

5.3.18 Maintenance of certification. Using the process flow chart and DESC Form EQM-42 as a baseline, the certified line will be monitored periodically by the certification team or its representatives. This audit is to assure that the line is still in certified condition and that all process and process control changes are properly documented and implemented.

* 5.3.19 Handling and storage. The procedures used for handling and storing ESD sensitive devices shall be specified.

* 5.3.20 Lid Torque Test. Glass frit sealed packages shall be subjected to the Lid Torque Test (Method 1018, MIL-STD-883). A sampling and test plan shall be submitted to the certification team for approval (e.g., one out of every 200 packages to be tested).

5.3.21 Marking. The procedures and materials used for marking shall be specified.

5.4 Changes from previous issue. The margins of this standard are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Air Force - 17
Army - ER
Navy - EC
NASA - NA

Review activities:
Air Force - 11, 19, 85, 99
Army - MI, AR
Navy - SH
DLA - ES

User activities:
Army - SM
Navy - CG, MC, AS, OS

Agent:
DLA - ES

Preparing activity:
NASA - NA

(Project 5962-0344)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

APPENDIX A

CROSS INDEX
OF CALIBRATION SYSTEM REQUIREMENTS
(MIL-STD-45662 VS. COMPANY DOCUMENTATION)

COMPANY NAME _____

ADDRESS _____

PREPARED BY _____

TITLE _____ DATE _____

Please indicate where each program element is covered in the company documentation. If documentation does not exist because a specific program element does not apply, so state and explain. If additional space is required, us a separate sheet. (May be handscribed.)

PROGRAM ELEMENT AND MIL-STD-45662 PARAGRAPH	PLEASE LIST COMPANY DOCUMENT TITLE, NUMBER REVISION, SECTION, PARAGRAPH, PAGE, ETC. (AS APPLICABLE)
CALIBRATION	
a. Description of Calibration System	
(1) Controls for segregation of obsolete, damaged or otherwise inaccurate equipment	
(2) Controls for verification of production tooling (jigs, fixtures, etc.) when used	
(3) List of Measurement Standards (reference and transfer) DESC Form 36	
(4) Availability of system description, procedures, and calibration reports	
b. Adequacy of Standards Measurement standards established for calibration of test and measuring equipment have the capabilities for the intended use (5.2)	
(1) Accuracy	
(2) Stability	
(3) Range	
c. Environment Controls (5.3)	
(1) Environmental conditions controlled to the extent neceesary to assure continued measurements of the required accuracy. Consideration given to:	
Temperature	
Humidity	
Vibration	
Cleanliness	
Other controllable factors affecting precise measurement	
(2) Controls for the application of compensating corrections to calibration results obtained in environments other than standard	
d. Calibration intervals (5.4) Procedures provide control for:	
(1) The calibration of measuring and test equipment and measurement standards at periodic intervals	
(2) The establishment of an interval period based upon stability, purpose, and degree of usage	

DESC Form 695
APR 81

Previous Edition can be used until exhausted

Page 1 of 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

APPENDIX A - Continued.

PROGRAM ELEMENT AND MIL-STD-45662 PARAGRAPH	PLEASE LIST COMPANY DOCUMENT TITLE, NUMBER, REVISION, SECTION, PARAGRAPH, PAGE, ETC (As Applicable)
(3) The adjustment of interval periods when evidenced by the results of previous calibrations	
e. Calibration Procedures (5.5)	
(1) Preparation, provision, and utilization of written procedures for the calibration of measuring and test equipment and measurement standards	
(2) Requirement for calibration to be performed by comparison with higher accuracy level standards	
(3) Utilization of published standard practices or manufacturers instructions	
(4) Surveillance or checks that procedures are being followed	
(5) Availability of procedures to calibration personnel	
f. Out of tolerance evaluators (5.6)	
(1) Evaluation of suspect products (5.6.1)	
(a) Procedures to determine impact of out of tolerance equipment on product quality	
(b) Corrective action procedure for out of tolerance equipment	
(c) Records of results of analysis and corrective action	
(d) Records are available to Government representative	
(2) Evaluation of Calibration system accuracy (5.6.2)	
(a) Procedures evaluate the calibration system based on results of generated calibration test data	
(b) Calibration system includes: Adjustment of calibration frequency, adequacy of measuring or test instrument, review of calibration procedure; identification of unsatisfactory or inaccurate equipment	
g. Calibration Sources (5.7)[1] Procedures provide control for:	
(1) The calibration of test and measuring equipment by a source whose standards are traceable to the National Bureau of Standards	
(2) The calibration of reference standards by a capable commercial facility, a Government laboratory or the National Bureau of Standards	
(3) A report, certificate, or data sheet attesting to the data, accuracy, and conditions under which the calibration results of reference standards were obtained	
(4) The provision of reports, record cards, etc., for subordinate standards, measuring and test equipment when such information is deemed essential	
(5) Assuring that calibration sources other than the National Bureau of Standards or a Government laboratory are in fact capable of performing the required service	
(6) Producing such reports for review by the Government representative	
h. Application and Records (5.8)[1] (5.2.5(c))[2] Procedures provide control for:	
(1) Supporting records to show that established schedules and procedures are applied to maintain the accuracy of measuring and test equipment and measurement standards	
(2) An individual record of calibration for each item of measuring and test equipment and measurement standard providing calibration interval, data of certification and result of last calibration	

DESC Form 695
APR 81

Page 2 of 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

APPENDIX A - Continued.

PROGRAM ELEMENT AND MIL-STD-45662 PARAGRAPH	PLEASE LIST COMPANY DOCUMENT TITLE, NUMBER REVISION, SECTION, PARAGRAPH, PAGE, ETC (As Applicable).
(3) Noting the report of certificate number on the individual record of these items whose accuracy is reported by calibration report or certificate	
i. Calibration status (5.9) Procedure provides control for:	
(1) The labeling or other means shall be established to assure adherence to calibration schedules. The system shall indicate the date of last calibration, by whom calibrated, and when the next calibration is due.	
(2) An identifying code to reflect the status of serviceability for those items whose size or functional characteristics prohibit the application of a label	
(3) The monitoring of recall records is assure adherence to calibration schedules	
(4) Labels, codes, or recall records indicating the applicable condition of those items which are not required to be used to their full capabilities or which require a functional check only	
j. Control of subcontractor calibration to this standard is assured by the contractor to the degree necessary to assure compliance with contractor requirements (5.10)	
k. All measuring and test equipment handling procedures provide assurance that transportation shall not adversely affect the equipment (5.11)	
l. Amendment and revisions (5.12)	
(1) The contractor may allow or authorize his subcontractor to follow the amended or revised standard subsequent to contractually effective date, where no increase in fee or price is involved	
(2) The contractor shall not be required to follow the amended or revised standard except as a change in the contract	
(3) Procedures provide for notification of the contracting officer	

DESC Form 695
APR 81

Page 3 of 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

CHECK APPLICABLE BLOCK ☐ LIST OF QUALIFICATIONS TEST FACILITIES

or

☐ LIST OF QUALITY CONFORMANCE TEST FACILITIES

☐ MASTER LIST ☐ SPECIFICATION LIST

NAME OF FACILITY _____

STREET _____

CITY _____ STATE _____

SPEC NO _____

AMEND _____

MIL-STD _____

LIST DATE _____

REVISED _____

SPEC. PAR OR TEST METHOD	EQUIPMENT	MANUFACTURER	TYPE OR MODEL	SERIAL OR INVENTORY NO.	DESCRIPTION AND USE (Include parameters to be measured, measuring devices and controls as applicable)	EQPT LIMITS (Include Multiple Ranges)	ACCU-RACY	FREQ OF CALIB

DESC FORM 36
SEP 71

PREVIOUS EDITION MAY BE USED

PAGE OF

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE: MIL-STD-976A Certification requirements for JAN microcircuits

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address – Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:18Z
Check the source to verify that this is the current version before use.